Canon Bruce

Rochester, Illinois
(217) 299-2658
canon.m.bruce@vanderbilt.edu

EDUCATION

Vanderbilt University — **Nashville, Tennessee**

Bachelor of Engineering, Majors in Electrical and Computer Engineering, Minors in Computer Science — ***May 2028***

GPA: 3.34/4.0

Relevant Coursework: Linear Algebra | Differential Equations | Data Structures | Intermediate Software Design

PROFESSIONAL EXPERIENCE

SiteVue AI — **Franklin, Tennessee**

Electrical Engineer Intern — ***June 2026 – Aug 2026***

- Collaborated with engineering leadership to develop autonomous robotics solutions and support deployment opportunities.
- Developed a Linux-based robotics platform integrating computer vision, AI decision-making, and precision arm control to advance SiteVue's R&D efforts in industrial automation.
- Designed a computer vision pipeline for an autonomous floor-painting robot capable of identifying, locating, and painting tiles.

Legacy Audio — **Springfield, Illinois**

Software Engineer Intern — ***March 2026 – May 2026***

- Architected Raspberry Pi-based Qt touchscreen interface, replacing legacy physical controls on Legacy Audio's in-house DSP unit
- Led end-to-end touchscreen integration efforts, optimizing display configuration, hardware interfacing, and deployment readiness for live product demonstration
- Contributed to the transition from outsourced to in-house DSP production, reducing per-unit costs from $2,500 to $1,200

Vision Care Associates — **Springfield, Illinois**

Lab Technician Intern — ***May 2024 – Aug 2025***

- Operated precision optical machinery to fulfill 500+ patient lens prescriptions, maintaining throughput with zero quality escapes
- Streamlined daily lab workflows by proactively communicating status updates to senior technicians, reducing delays by 20%
- Cross-trained across multiple stages of the lens fabrication process, increasing operational flexibility within the lab

Elks Lodge — **Springfield, Illinois**

Head Lifeguard — ***May 2022 – Aug 2023***

- Managed schedules and shift coverage for a team of 12 lifeguards, ensuring zero understaffed shifts across full summer season
- Implemented recurring CPR and first aid training sessions every 2-3 weeks, increasing emergency preparedness

LEADERSHIP & EXTRACURRICULAR ACTIVITIES

Vanderbilt Quantum Initiative — **Nashville, Tennessee**

Research Associate — ***Jan 2025 – Present***

- Studying quantum communication principles as part of a 40-member interdisciplinary research initiative at Vanderbilt
- Contributing to satellite simulation project exploring implementation of quantum communication in low-earth orbit systems
- Synthesized findings from academic literature and industry research to support project direction and technical decision-making

Theta Tau, Professional Engineering Fraternity, Vanderbilt University — **Nashville, Tennessee**

Intramural Chair — ***Aug 2025 – Present***

- Coordinate intramural programming across 6 teams and 5 sports for a 100+ member professional engineering fraternity
- Introduced a points-based reward system tying event attendance and performance to a semester-end prize, driving stronger team spirit and engagement across intramural programming
- Grew chapter intramural participation from 30% to 50% by restructuring team rosters and streamlining scheduling processes

ECE 2212, Digital Systems, Vanderbilt University — **Nashville, Tennessee**

Teaching Assistant — ***Aug 2026 – Present***

- Lead weekly lab sections for over 40 students, guiding hands-on hardware debugging and SystemVerilog implementation
- Run group study sessions each week, helping 20+ students work through digital logic design problems and prepare for exams

TECHNICAL SKILLS

Python, C++, Java, Claude Code, Computer Vision, MATLAB, Microsoft Excel, Git, LTspice, Quartus

INTERESTS

Fitness & Human Performance, NBA Analytics, Virtual Reality, Venture & Startups, Pokémon, SEC Basketball & Football